

Project Title	Daily Transactions
Tools	Visual Studio code / jupyter notebook
Domain	Finance Analyst
Project Difficulties level	intermediate

About Dataset

The "Daily Transactions" dataset contains information on dummy transactions made by an individual on a daily basis. The dataset includes data on the products that were purchased, the amount spent on each product, the date and time of each transaction, the payment mode of each transaction, and the source of each record (Expense/Income).

This dataset can be used to analyze purchasing behavior and money management, forecasting expenses, and optimizing savings and budgeting strategies. The dataset is well-suited for data analysis and machine learning applications, it can be used to train predictive models and make data-driven decisions.

Column Descriptors

- **Date:** The date and time when the transaction was made
- **Mode:** The payment mode used for the transaction
- **Category:** Each record is divided into a set of categories of transactions
- **Subcategory:** Categories are further broken down into Subcategories of transactions
- **Note:** A brief description of the transaction made
- **Amount:** The transactional amount
- **Income/Expense:** The indicator of each transaction representing either expense or income
- **Currency:** All transactions are recorded in official currency of India

1. Project Review:

Objective:

- Analyze daily financial transactions to identify trends, patterns, and insights.
- Generate a comprehensive report with visualizations.

2. Dataset Description

- **Date:** Date of the transaction.
- **Transaction_ID:** Unique identifier for each transaction.
- **Account_ID:** Unique identifier for the account.
- **Category:** Category of the transaction (e.g., Sales, Purchase, Salary).
- **Amount:** Amount of money involved in the transaction.
- **Type:** Type of transaction (Credit or Debit).

3. Steps to Complete the Project

Step 1: Import Libraries and Load Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the dataset
df = pd.read_csv('daily_transactions.csv')

# Display the first few rows of the
dataset df.head()
```

Step 2: Data Cleaning

- Handle missing values.
- Correct data types.
- Remove duplicates.

```
# Check for missing
values df.isnull().sum()

# Fill or drop missing values df['Category'].fillna('Unknown',
inplace=True) df.dropna(subset=['Date', 'Transaction_ID',
'Amount'], inplace=True)

# Convert data types
df['Date'] = pd.to_datetime(df['Date'])
df['Amount'] = df['Amount'].astype(float)

# Remove duplicates
df.drop_duplicates(inplace=True)
```

```
# Verify data  
types df.dtypes
```

Step 3: Exploratory Data Analysis (EDA)

- Summary statistics.
- Distribution of transaction amounts.
- Transaction counts by category and type.

```
# Summary statistics  
df.describe()  
  
# Distribution of transaction amounts  
plt.figure(figsize=(10, 6))  
sns.histplot(df['Amount'], bins=50, kde=True)  
plt.title('Distribution of Transaction Amounts')  
plt.xlabel('Amount') plt.ylabel('Frequency')  
  
plt.show()  
  
# Transaction counts by category  
plt.figure(figsize=(12, 6))  
sns.countplot(data=df, x='Category', order=df['Category'].value_counts().index)  
plt.title('Transaction Counts by Category')
```

```
plt.xlabel('Category')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()

# Transaction counts by type
plt.figure(figsize=(10, 6))
sns.countplot(data=df, x='Type')
plt.title('Transaction Counts by
Type') plt.xlabel('Type')
plt.ylabel('Count')
plt.show()
```

Step 4: Time Series Analysis

- Trend analysis.
- Monthly and daily trends.

```
# Resample data to monthly frequency
monthly_data = df.resample('M', on='Date').sum()

plt.figure(figsize=(14, 7))
plt.plot(monthly_data.index, monthly_data['Amount'], marker='o')
plt.title('Monthly Transaction Amounts')
plt.xlabel('Month')
plt.ylabel('Total Amount')
```

```
plt.grid(True)
plt.show()

# Daily trends
daily_data = df.groupby(df['Date'].dt.date).sum()

plt.figure(figsize=(14, 7))
plt.plot(daily_data.index, daily_data['Amount'], marker='o')
plt.title('Daily Transaction Amounts')
plt.xlabel('Date')
plt.ylabel('Total Amount')
plt.grid(True)
plt.show()
```

Step 5: Correlation Analysis

- Analyze the correlation between transaction categories and amounts.

```
# Create a pivot table for correlation analysis
pivot_table = df.pivot_table(index='Date', columns='Category',
values='Amount', aggfunc='sum', fill_value=0)

# Calculate correlation matrix
correlation_matrix = pivot_table.corr()
```

```
# Plot correlation heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm',
linewidths=0.5) plt.title('Correlation Heatmap of Transaction Categories')
plt.show()
```

Step 6: Generate Report

- Summarize findings and visualizations.

4. Report

Summary

The financial transactions dataset was analyzed to identify key trends and insights. The data cleaning process involved handling missing values, correcting data types, and removing duplicates. Exploratory Data Analysis (EDA) revealed the distribution of transaction amounts, transaction counts by category and type, and significant patterns over time. Time series analysis highlighted monthly and daily transaction trends. Correlation analysis identified relationships between different transaction categories.

Key Findings

- The distribution of transaction amounts showed a right-skewed pattern with most transactions clustered around lower values.
- Sales and Purchases were the most common transaction categories.
- Credit transactions were more frequent than Debit transactions.
- Monthly transaction trends revealed seasonal patterns with peaks in certain months.
- Correlation analysis indicated strong relationships between certain transaction categories.

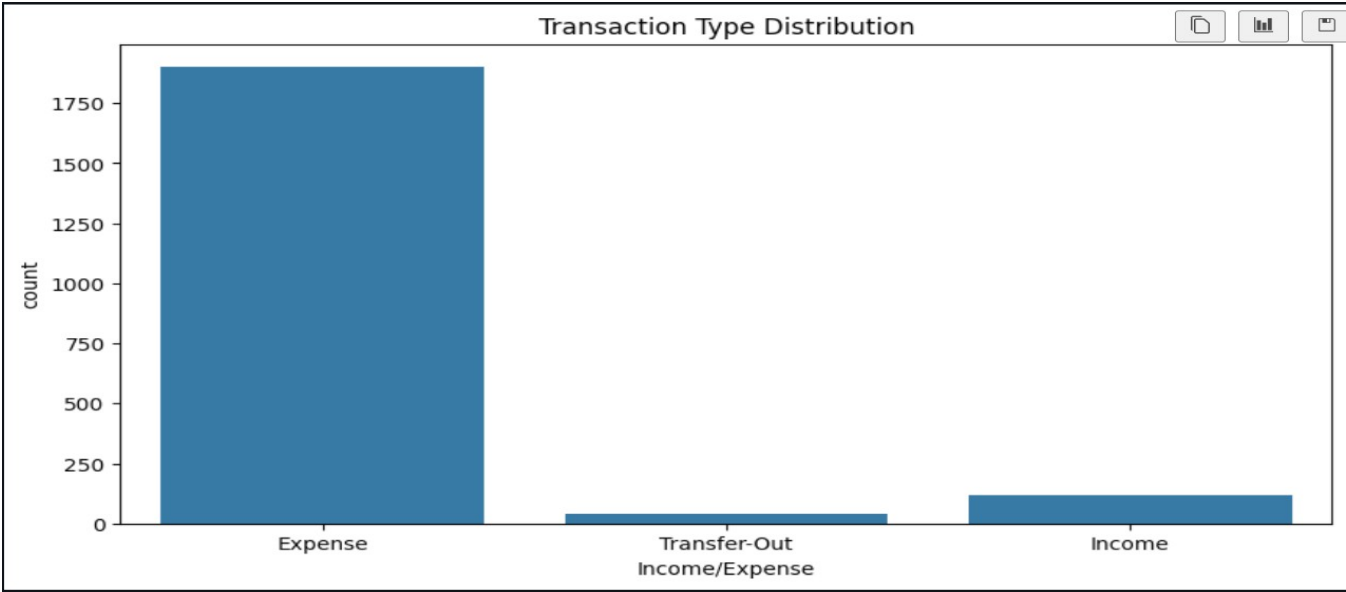
Visualizations

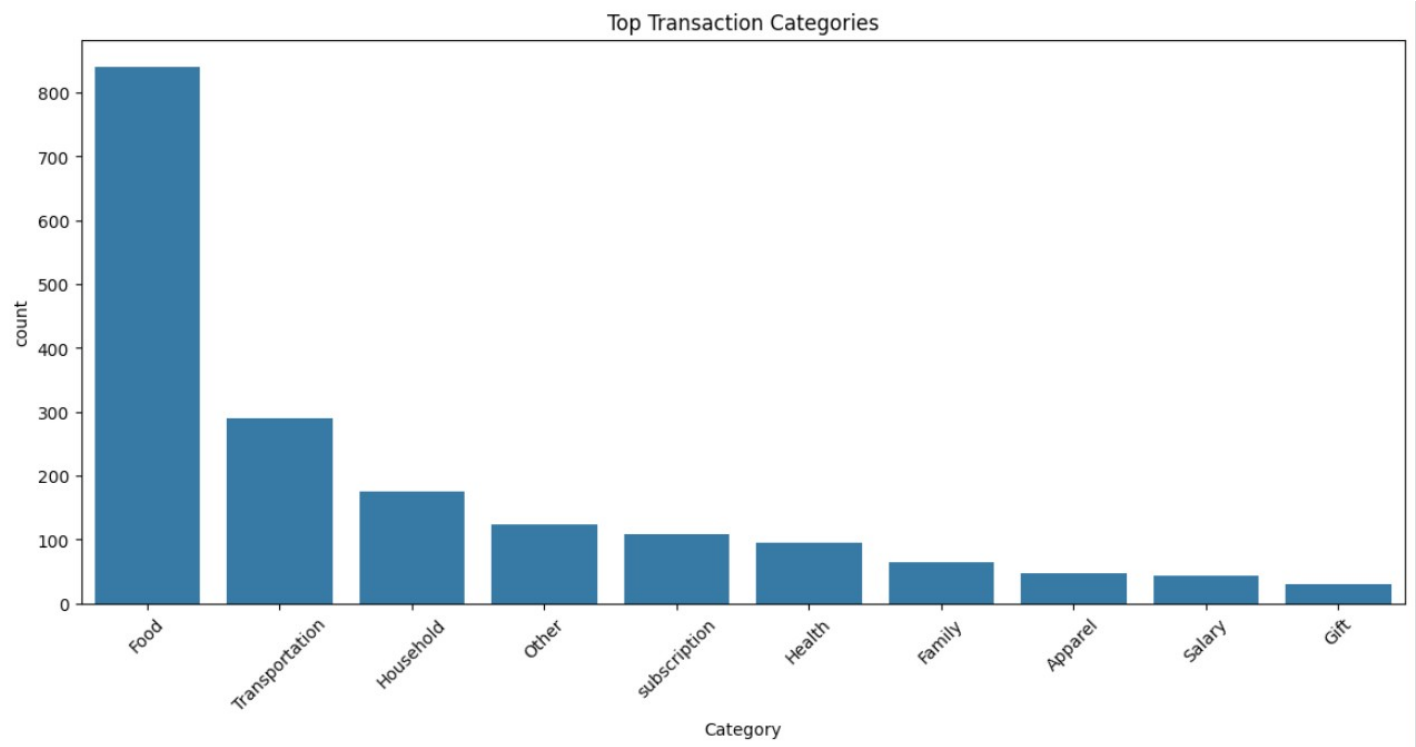
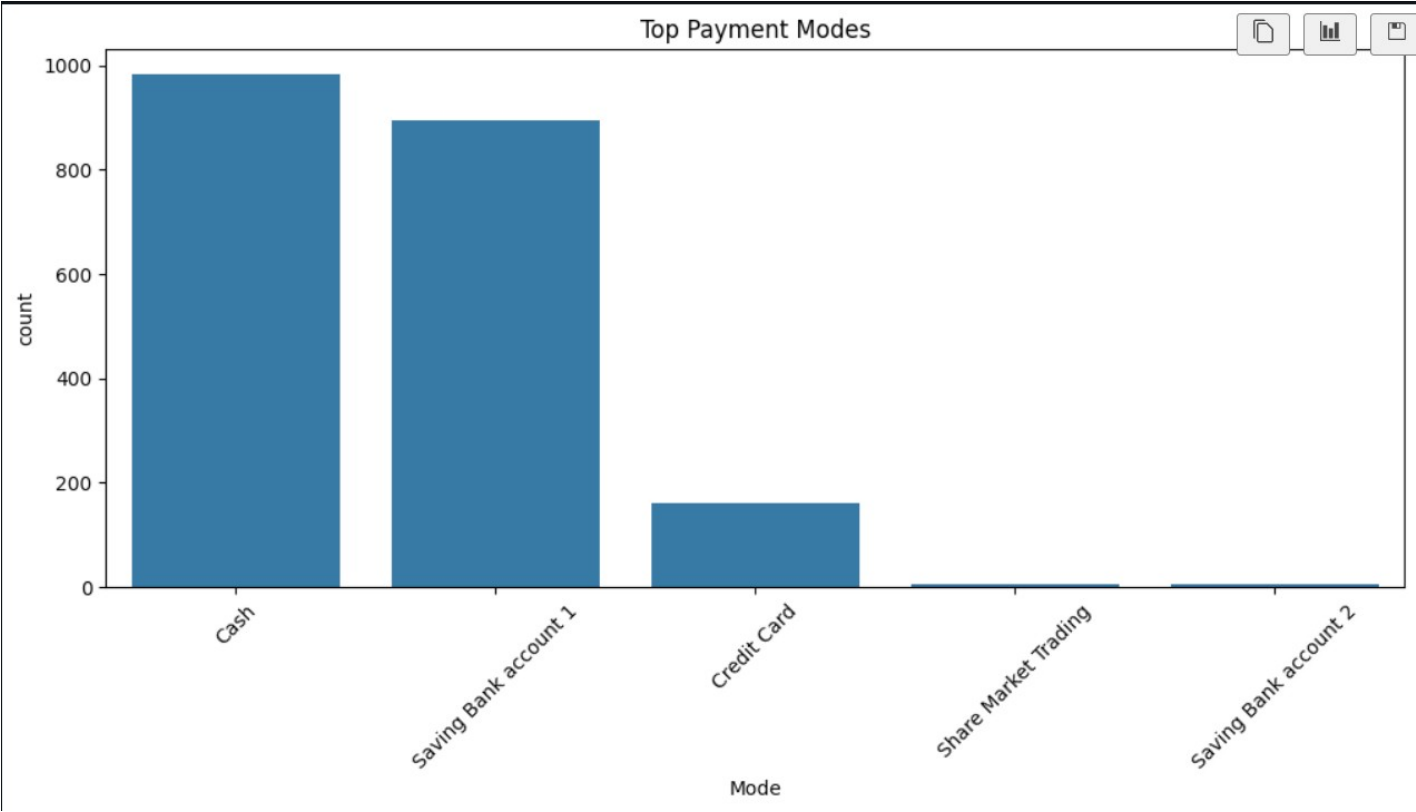
- Distribution of Transaction Amounts
- Transaction Counts by Category and Type
- Monthly and Daily Transaction Amounts
- Correlation Heatmap of Transaction Categories

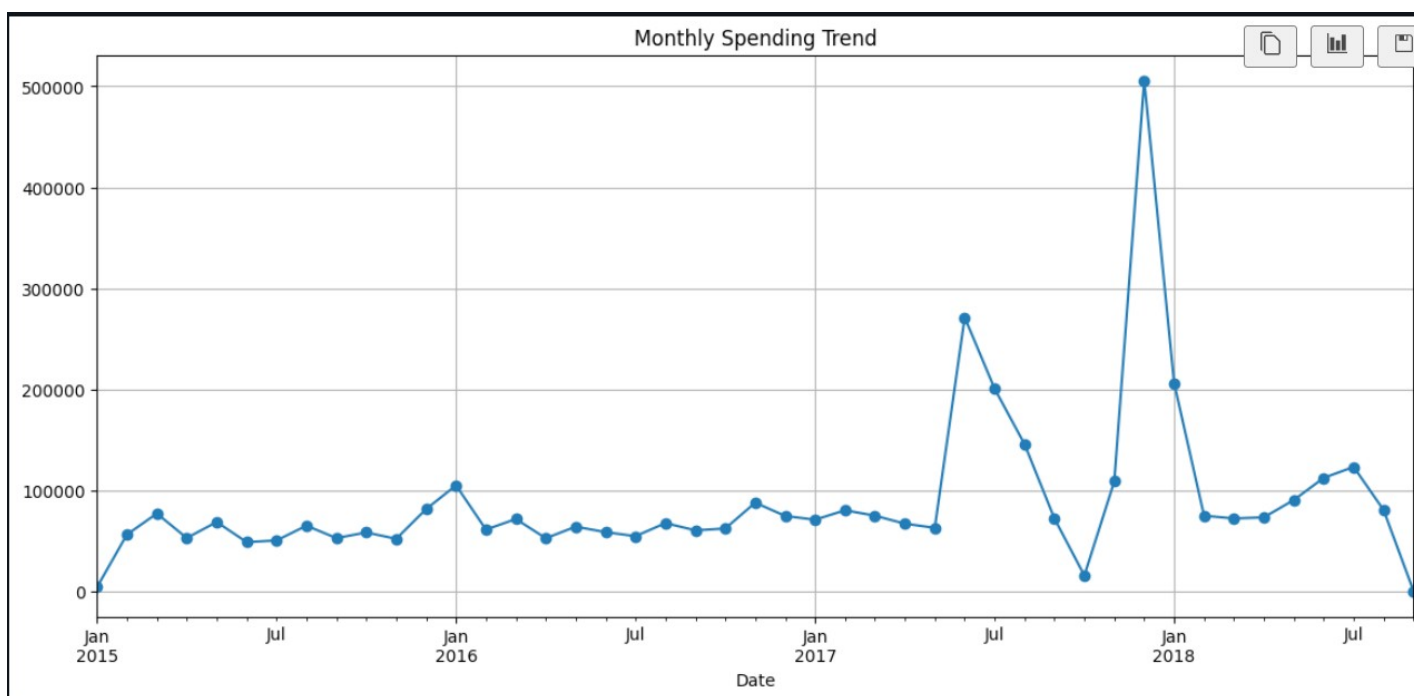
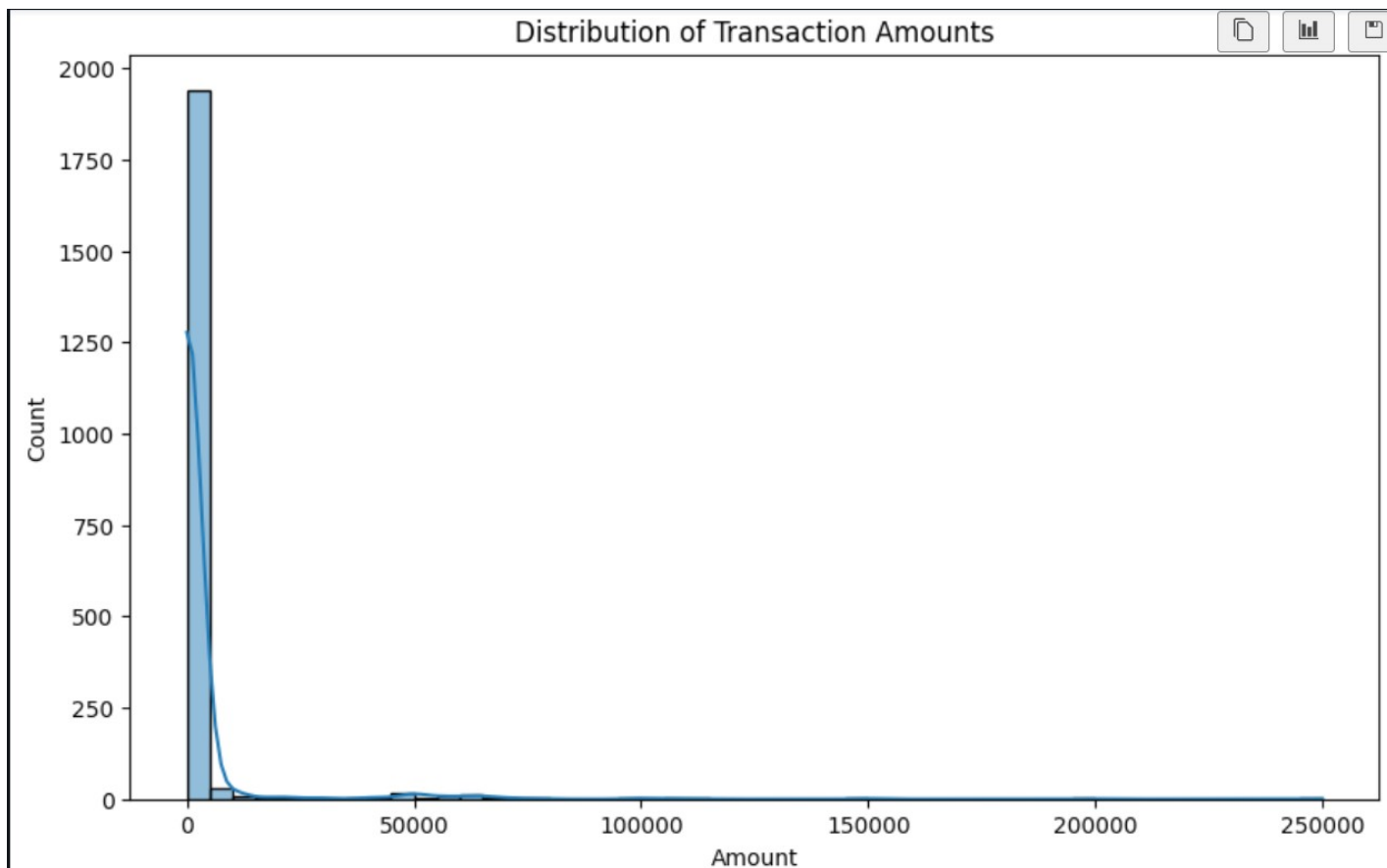
This project provides valuable insights into daily financial transactions, helping to inform decision-making and strategic planning.

Output Pictures:

	Date	Mode	Category	Subcategory	Note	Amount	Income/Expense	Currency
0	20/09/2018 12:04:08	Cash	Transportation	Train	2 Place 5 to Place 0	30.0	Expense	INR
1	20/09/2018 12:03:15	Cash	Food	snacks	Idli medu Vada mix 2 plates	60.0	Expense	INR
2	19/09/2018	Saving Bank account 1	subscription	Netflix	1 month subscription	199.0	Expense	INR
3	17/09/2018 23:41:17	Saving Bank account 1	subscription	Mobile Service Provider	Data booster pack	19.0	Expense	INR
4	16/09/2018 17:15:08	Cash	Festivals	Ganesh Pujan	Ganesh idol	251.0	Expense	INR







Daily Spending Trend

